Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18IOT7DE41]

SEVENTH SEMESTER B. Tech EXAMINATION, DECEMBER-2021

MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Explain the term 'well-posed learning problems' in the context of machine learning and further about its three features as Learning task, Performance measure, and Training experience in brief. **07**

b. Mention at least five common issues in machine learning and explain about them in short. **07**

2. a. Demonstrate the data preprocessing technique and further at least four aspects of requirement of data preprocessing. **07**

b. Describe the following data pre-processing methods in brief: **07**

i. Data Cleaning
ii. Data Integration
iii. Data Transformation
iv. Data Reduction

3. a. Delineate the need of applying regression and further mention the conditions when to apply Simple Linear Regression, Multiple Linear Regression and Polynomial Regression. **07**

b. Calculate the Root Mean Square Error (RMSE) for the following data: **07**

Actual	12	16	14	12	15	18	16
Predicted	15	16	18	18	14	18	15

OR

4. a. Explain the following terms: **07**

i. Root Mean Square Error (RMSE)
ii. Mean Absolute Error
iii. Correlation

b. Compute the Pearson Correlation Coefficient for the following data: **07**

Length	12	14	14	13	15	18	16
Width	15	18	15	18	14	16	15

5. a. Write the steps for applying decision tree on any given data. Moreover explain the following terms with example: **07**

   i. Entropy

   ii. Information gain

   iii. Gini index

   b. Describe the following algorithms in brief: **07**

   i. Support Vector Machine (SVM)

   ii. K-Nearest Neighbours (K-NN)

OR

6. a. Differentiate among regression, classification and clustering with appropriate numerical examples. **07**

   b. Explain the following algorithms in brief: **07**

   i. Random forest classification

   ii. Naïve Bayes algorithm

7. a. Write about the association rule mining and how it is different from clustering. **07**

   b. Demonstrate the general architecture of artificial neural network model and further explain its functional similarity with human mind. **07**

OR

8. a. Explain about apriori algorithm with at least two examples. **07**

   b. Elucidate the functionality of genetic algorithm in machine learning including the explanation about Selection, Recombination, Mutation and Elitism. **07**